

Project 19

Apparent Barriers in iiwa 7 Redundancy Charts

Format: Group of two students; simulation-based project.

Course foundations: Lecture 8; Exercise 4.

Motivation

Fixing a redundant joint converts a 7R IK problem into a square subproblem, but that subproblem can become singular while the full robot remains regular. A planner may then report an apparent barrier caused by its chosen coordinate. A local change of coordinate can reveal whether the underlying robot motion continues.

Task

Overall objectives. Analyze a supplied iiwa path with a fixed- q_3 chart failure and determine whether a second redundancy coordinate can continue the same full-pose IK family without violating robot constraints.

- **Reproduce the supplied case.** Use the instructor-provided short pose path, initial IK, and suspected chart-fold interval. Validate the iiwa URDF/D–H model and reproduce the failure of fixed- q_3 continuation with documented residuals, limits, and solver tolerances.
- **Separate the two regularity questions.** Compare $\sigma_{\min}(J)$ for the scaled full 6×7 Jacobian with $\det \bar{J}_3$, where $\bar{J}_3$ removes column 3. Explain how a singular reduced matrix can reflect tangency of the self-motion family to the chosen coordinate rather than rank loss of the full robot.
- **Construct a second local chart.** Select one alternative joint q_j for which the corresponding six-column Jacobian is well conditioned near the supplied interval. Hold q_j fixed in a local numerical corrector and continue the same recovered configuration family. Use only this one alternative chart and a bounded overlap region.
- **Classify and reconcile the charts.** Generate path-sample versus redundant-angle panels for q_3 and for q_j . Preserve the supplied fixed- q_3 labels and assign traceable local continuation identities in the second chart. Match overlapping solutions by full joint configuration and pose closure, not by equal labels or determinant signs alone.
- **Validate the crossing or obstruction.** Attempt one trajectory using the chart overlap and compare with the original method. Check intermediate poses, full-Jacobian regularity, limits, and continuity. Mark which chart barrier was removed and which failures persist; failure of two charts alone does not prove robot infeasibility.

Specifics and scope

The instructor will supply the path, starting solution, and a known candidate fold interval. Limit the study to two coordinates, one overlap, and one local chart transition, with roughly 80 path samples and 100 angle samples per chart initially. Reuse the fixed- q_3 analytical IK; a local numerical corrector is sufficient for the alternative coordinate.

For every chart, keep pose validity, joint limits, full-Jacobian regularity, and reduced-chart regularity as separate diagnostics. The rectangular 6×7 Jacobian has no determinant. A feasible pixel is not a validated edge: re-solve intermediate prescribed poses and check joint continuity within limits. If collision checking is unavailable, state that feasibility is kinematic only; a clearance proxy is not a full collision test.

Expected outcome

Understand coordinate-induced IK ambiguity and distinguish it from a physical singularity or constraint. Deliver two classified feasibility charts, overlap matches, full/reduced regularity plots, and a validated continuation or an explicit account of the unresolved obstruction.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The KUKA LBR iiwa 7 URDF is `lectures_main/assets/models/iiwa7/iiwa7.urdf`; STL meshes are in that directory. The accompanying `iiwa7_free_joints.urdf` is a mathematical continuation model; apply physical limits separately. The course analytical IK model will be supplied; reuse it rather than deriving a general 7R solver. Lecture 8 explains fixed-redundancy slices, self-motion charts, and reduced versus full Jacobians. The special path and fold seed will be supplied separately; the reference path CSV alone is not guaranteed to contain the required event.

Use the lecture website to verify D–H parameters, visualize IKs and singularities, and simulate paths. All listed paths are relative to the repository root. Start the website following `lectures_main/README.md`.

Workload and collaboration

Budget approximately 60 hours per student (120 person-hours per pair) for this 2-ECTS project, following EPFL’s 30-hours-per-credit convention.¹ Deduct any lectures or exercises included in those credits. Allocate roughly 20 team-hours to model validation, 50 to the core work, 30 to experiments, and 20 to reporting. Reuse the specified IK and simulations. Share validation and interpretation even if modelling and implementation are divided between students. Hardware execution is outside scope.

¹EPFL: Design a Course, workload guidance.